

Introduction to Data Science

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1 Regression Numericals

Given Dataset:

Hours Exercised (x) : [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

Change in Weight (y) : [0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0]

Step 1: Calculate the Mean of x and y We calculate the mean of x (hours exercised) and y (change in weight):

$$\bar{x} = \frac{1}{10} \sum_{i=1}^{10} x_i \bar{y} = \frac{1}{10} \sum_{i=1}^{10} y_i$$

For x :

$$\bar{x} = \frac{1}{10}(1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10) = \frac{1}{10} \times 55 = 5.5$$

For y :

$$\bar{y} = \frac{1}{10}(0.5 + 1.0 + 1.5 + 2.0 + 2.5 + 3.0 + 3.5 + 4.0 + 4.5 + 5.0) = \frac{1}{10} \times 25 = 2.5$$

So, $\bar{x} = 5.5$ and $\bar{y} = 2.5$.

Step 2: Calculate the Slope (m) and Y-Intercept (b) The slope (m) and y-intercept (b) of the regression line can be calculated using the formulas:

$$m = \frac{\sum_{i=1}^{10} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{10} (x_i - \bar{x})^2} b = \bar{y} - m \cdot \bar{x}$$

First, let's calculate m :

$$m = \frac{\sum_{i=1}^{10}(x_i - 5.5)(y_i - 2.5)}{\sum_{i=1}^{10}(x_i - 5.5)^2} = \frac{(1 - 5.5)(0.5 - 2.5) + (2 - 5.5)(1.0 - 2.5) + \dots + (10 - 5.5)(5.0 - 2.5)}{(1 - 5.5)^2 + (2 - 5.5)^2 + \dots + (10 - 5.5)^2} = \frac{(-4.5)(-2.0) + (-3.5)(-1.5) + \dots + (-5.5)(-2.5)}{16 + 9 + \dots + 1} = \frac{9.0 + 5.25 + \dots + 13.75}{55} = \frac{82.25}{55} \approx 1.49545$$

Now, let's calculate b :

$$b = 2.5 - 0.8636 \times 5.5 \approx 2.5 - 4.7499 \approx -2.2499$$

So, $m \approx 0.8636$ and $b \approx -2.2499$.

Step 4: Make Predictions Once we have m and b , we can make predictions for $x = 9.5$ using the equation of a straight line:

$$\hat{y} = mx + b$$

Let's calculate $\hat{y}$:

$$\hat{y} = 0.8636 \times 9.5 - 2.2499 \approx 8.2034 - 2.2499 \approx 5.9535$$

So, the predicted change in weight for $x = 9.5$ hours exercised is approximately 5.9535 kilograms.

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